

legoESM ocean — simulation choice worksheet

Pick one option per axis. Print and mark up.

Status legend — production default · available, validated · caveat / less tested · avoid (broken or not wired) · planned but not implemented

Each row below is an independent axis of choice. Setting up a run = pick one leaf per row. Cross-axis restrictions are annotated. “Invalid / unwired combinations” are listed at the end.

1 · Grid

GRID

latlon C-grid
 regular (uniform $\Delta \times \Delta$)
 Mercator (isotropic-area conformal; `docs/mercator\_grid\_plan.md`)
subtypes:
 global
 regional (rectangular subdomain, e.g. 24×48)
 channel (zonally periodic band, e.g. 24×72, 200×100, 20×10, 20×18)

MPAS Voronoi C-grid (TRiSK)
 global icosahedral (ico3 300 km, ico4 150 km, ico5 stiff)
 regional (300 km Voronoi patch)
 channel (300 km / 100 km / 10 km)

cubed-sphere C-grid (C24, C48 ...)
 face-boundary instability (project_cubesphere_ocean_instability)
 - avoid in production until fixed

spectral / Gaussian (T21)
 standalone runner only (`scripts/run\_ocean\_spectral\_tests.py`);

not wired into the main matrix

2 · Resolution

RESOLUTION (typical - what the test matrix uses)

latlon

global:	36×72 (~2.5°)	48×72	1/4° + finer	
regional:	24×48			
channel:	24×72	200×100 (10 km)	20×18	20×10

MPAS

global:	ico3 (~300 km)	ico4 (~150 km)	ico5
regional:	300 km		
channel:	300 km	100 km	10 km

cubed-sphere: C24 (higher untested)

3 · Vertical coordinate

VERTICAL

z* (z-star, Adcroft & Campin 2004)
with partial bottom cells
without (full-step bathymetry)

ALE remap onto z* (planned; MOM6-style)
isopycnal / hybrid (not started)
/ terrain-following (not started)

Layer count `n_levels`: typically 10 (idealized) or 50–75 (production-style). Default `H_max` = 5500 m. Single-layer (`nlev=1`) supported for truly-barotropic runs.

4 · Time integration — per term, not one axis

The ocean dycore does **not** use a single “outer integrator”. Different terms have different stiffness and oscillation character and each one is stepped with the scheme suited to it. The table below documents what is actually in `_step_impl` today.

Term	Lat-lon (current)	MPAS (current)	Reference	Stability character
Planetary Coriolis $f \times \mathbf{u}$	forward-backward (Matsuno)	forward-backward (Matsuno)	Matsuno; Hallberg 97	neutral, $\ G\ ^2 = 1 + O((f\Delta t)^4)$
Relative-vorticity / PV flux	AL81 spatial in outer step	TRiSK <code>pv_scheme</code> (default "enstrophy")	Arakawa-Lamb 81; Ringler 10	conservative; spatial choice sets KE/enstrophy budget
Horizontal pressure gradient	within outer step (Euler)	within outer step (Euler)	Adcroft-Campin 04	hydrostatic; no own stepper
Momentum advection (vec-inv)	vector-invariant in outer step (Euler)	TRiSK vector-invariant in outer step	Sadourny 75	advection-CFL
Tracer advection	<code>tracer_time_integrator</code> { <code>euler()</code> , <code>ab2</code> , <code>rk3</code> }	implicit in outer step	Hundsdoerfer 95	CFL 1 (Euler), 0.72 (AB2), 1 (SSP-RK3)
Barotropic subcycle	explicit BEBT pred-corr (or implicit-CN)	explicit subcycle (or implicit-CN)	SMC05; Hallberg 97	gravity-wave CFL on subcycle; CN unconditional
Vertical mixing (diffusion+KPP)	within outer step (currently explicit — needs audit per item 5C of plan-of-work)	same	LMD 94; CVMix	$\Delta t < \Delta z^2 / (2K_v)$ if explicit; unconditional if implicit

`dt_outer:` 300 s @ ~2.5° (advection-CFL limited)
`n_barotropic_substeps:` ~20-60 (gravity-wave-CFL limited if explicit)

The `src/legoesm/timestepping/dispatch.py` module exposes Forward Euler, Heun/RK2, SSP-RK3, SSP-RK34, RK4 integrators but the ocean dycore does **not** route through it — split-explicit per-term stepping above is the production pattern. The dispatch module is used by the atmosphere SW path.

5 · Mode splitting (barotropic baroclinic)

BAROTROPIC SOLVER

```

explicit subcycling
  time integrator
    BEBT predictor-corrector (Shchepetkin-McWilliams 2005)
    forward-backward (Hallberg 1997)
  time filter (over substeps)
    cosine window (ROMS/MOM6 style)
```

```

    boxcar
    none (aliasing risk)
slow-forcing handling: included (fixes mode-splitting noise)
MAXVEL clip: on (safety net)
implicit bottom-drag factor: on

```

```

implicit Crank-Nicolson (MPAS only - production for multi-year stability)
`barotropic_solver="implicit_cn"`
- fixes TRiSK null-branch growth (project_mpas_barotropic_noise, 5-yr OK)

```

Cross-axis note — explicit subcycling is the only option on lat-lon today. Implicit-CN is MPAS-only.

6 · Pressure gradient force

PGF

```

lat-lon C-grid
z* standard (column-integrated hydrostatic, EOS-pressure 2-pass)

```

MPAS

```

density-Jacobian (docs/ocean/density_jacobian_pgf_*.md)
  Shchepetkin-McWilliams 2003 variant
  alternative finite-volume form
z* standard

```

cubed-sphere

```

z* standard (contributes to face-boundary instability)

```

EOS-pressure iteration: 2-pass (iterate_eos_and_pressure_anomaly).

7 · Equation of state

EOS (ocean/eos.py)

linear $_T (T - T_{\text{ref}}) + _S (S - S_{\text{ref}})$	← idealized experiments
Roquet-2015 polynomial	← Wright-style nonlinear
full nonlinear (T,S,p)	← thermobaric/cabbeling-aware

Constants: $\rho_0 = 1025 \text{ kg/m}^3$, $T_{\text{freeze_ocean}} = 271.35 \text{ K}$ (intentional vs. freshwater 273.15 K).

8 · Coriolis / PV-flux discretization

Spatial scheme — how ζ flux and $f \times \mathbf{u}$ are discretised on the grid:

CORIOLIS / PV-FLUX (spatial)

```
lat-lon C-grid
  AL81 (Arakawa-Lamb 1981, energy + enstrophy conserving)
    with Le Sommer 2009 / Stewart-Dellar 2016 partial-cell weights
    (`pv_flux_al81_partial_cell`)
  KE gradient on the C-grid:
    centered (default; legacy bit-exactness)
    + Hollingsworth correction (commit d0183817)
      required for high-Ro flow, fixes spurious vortex stretching
  (Sadourny 4-point average is used only for the planetary-f
    Matsuno step - see temporal column in §4)
```

```
MPAS Voronoi (TRiSK)
  "enstrophy" (Ringler 2010 enstrophy-conserving) ← current default
  "energy"    (Ringler 2010 energy-conserving)
  "mixed"     ( blend, `pv_alpha`)
  AL81        (not in dispatch today - Phase 2D candidate)
```

```
APVM dissipation (PV upwind):
  on  (`apvm_dt`)
  off
```

Temporal scheme — how the planetary $f \times \mathbf{u}$ is stepped in time:

PLANETARY CORIOLIS (temporal)

forward-backward Matsuno (both grids today)

```
lat-lon: `_forward_backward_coriolis_3d` in
  `ocean_model_latlon_cgrid.py`
MPAS:    `_forward_backward_coriolis_mpas_3d` in
  `ocean_model_mpas.py`
```

Crank-Nicolson on rotation (deferred - Phase 5D)
exact discrete-KE conservation; needs 2×2 per-cell linear
solve; deferred until inertial-oscillation test reveals
Matsuno as a bottleneck (not expected at typical ocean $f\Delta t$).

Known issue — MPAS Coriolis double-counting on 10-level runs (issue #103). MPAS PV-flux default "**enstrophy**" is the opposite of Alistair's recommendation; Phase 2D switches it to "**energy**".

9 · Momentum advection

MOMENTUM ADVECTION

lat-lon C-grid
vector-invariant (Sadourny + KE-gradient + PV-flux)
WENO5-ILES (Silvestri et al. 2024, rotational WENO)
WENO7-ILES
D-term: on (also has off flag)

MPAS
TRISK PV-flux + KE gradient (vector-invariant only)

10 · Tracer advection

TRACER ADVECTION (thickness-weighted, flux form)

lat-lon C-grid (LatLonCGridOceanConfig.tracer_advection)
TVD (Van Leer; 2nd-order; default today)
upwind (1st-order; fallback)
DST-3 (3rd-order; Hundsdorfer 95 / Easter 93)
d0 cap; full benefit at CFL > 0.1 or RK3 (project_dst3_advection)
PPM (Colella-Woodward 84)
PPM_FCT (PPM + Zalesak FCT limiter)
DST-3 multidim (known unstable for strong flows)
WENO-5, WENO-7 (Silvestri 2024)
SOM (Prather 86; 9-moment; fully differentiable, no limiter)
→ for low-mixing cases (Eady, ACC; project_som_advection)

```
MPAS Voronoi (MPASOceanConfig.tracer_advection)
  TVD (Van Leer; production)
  upwind (1st-order; dataclass default)
  - no 3rd-order options wired on Voronoi today
```

Tracer time integrator (sub-step within outer step): Euler (lat-lon default) · AB2 · SSP-RK3

11 · GM / Redi (mesoscale eddy parameterization)

GM / REDI

```
off (default for eddy-resolving runs)
```

```
lat-lon C-grid (project_gm_redi_latlon)
  centered slope (`slope_scheme="centered"`)
  Griffies-1998 triads (`slope_scheme="triads"`)
modes:
  GM only (advective skew flux only)
  Redi only (isopycnal diffusion only)
  GM + Redi (combined)
```

```
MPAS Voronoi
  centered slope
  triads (Phase 5 of plan - not yet)
```

Slope tapering: smooth (sigmoid), not hard-clipped — required for AD.

12 · Horizontal dissipation

HORIZONTAL DISSIPATION

```
Laplacian (constant)
  momentum: A_h ( ~1e3-1e4 m2/s @ 1°)
  tracer: K_h ( ~1e3 m2/s)
```

```
Biharmonic (constant)
  momentum: B_h ( ~1e10-1e12 m/s)
```

-form: K__bih (MPAS only - selectively damps vorticity)
 Smagorinsky (flow-dependent; C_smag 2.0 for eddying)
 Leith / modified Leith (MPAS - vorticity-based, target enstrophy)
 Hyperdiffusion (Δ^2 , Δ^3 orders configurable)
 Barotropic divergence damping (targets grid-scale compressible modes)
 APVM (MPAS only - PV upwind dissipation; see §8)
Known issue — MPAS + ETOPO bottom-trapped instability is an algorithm-class limit; damping suite exhausted (project_mpas_etopo_instability). 90-day stable, multi-year requires algorithmic change.

13 · Vertical mixing

VERTICAL MIXING

KPP (Large-McWilliams-Doney 1994)
 on for production
 off for idealized

Convective adjustment
 enhanced K_v when $N^2 < 0$
 plume scheme (ocean/physics/convection/plume.py)
 off

Background diffusivity
 A_v (~1e-4 m²/s)
 K_v (~1e-5 m²/s)

14 · Bottom drag

BOTTOM DRAG

quadratic _b = _0 C_d |u_b| u_b C_d 2.5e-3

```
linear      _b = _0 r u_b      r    1e-4 m/s
free-slip   (idealized, no drag)
implicit treatment in barotropic mode:  on
```

15 · Surface forcing slots

FORCING (ocean-side interfaces only)

```
Wind stress (_x, _y)
  analytical (cos / sin2 / channel jet)
  field input
```

```
Heat flux (Q_net)
  + shortwave penetration (Q_sr split, commit 50d65485)
```

```
Freshwater flux (E-P)
  virtual salt flux (apply_freshwater_virtual_salt_top)
  real volume change
```

```
T / S restoring
  point-wise scalar T*, S*
  3-D arbitrary array T*(x,y,z), S*(x,y,z) (commit 50d65485)
  time integrator: implicit Euler (explicit)
  timescale: ~ 10-60 d (sponges ~ 1-5 d)
```

```
Sponge regions (regional / channel grids)
  T, S
  u, v
```

16 · Initial conditions

INITIAL STATE (init_*.py + experiments/)

```
Rest state (gravity-wave / PGF tests)
  stratified, with land
  stratified, no land
```

```

uniform T/S, with land
uniform T/S, no land

Wave / adjustment tests
  barotropic_wave           (gravity wave; res-matched grids)
  inertia_gravity_wave      (Bishnu et al. 2024)
  geostrophic_adjustment

Wind-driven gyres
  barotropic_double_gyre (cos & sin2 wind)
  baroclinic_gyre (cos & sin2)
  global_barotropic_wind (lat-lon, MPAS; cube )
  global_barotropic_wind_1lev

Density-current benchmarks
  lock_exchange (Ilicak protocol; lat-lon, cube only)
  overflow      (Ilicak; lat-lon, cube only)

Baroclinic instability
  phillips_two_layer (all grids)
  eady_instability   (channel, sponge)
  eady_uniform        (Bishnu - classical Eady)
  eady_gm_redi        (lat-lon centred + triads, MPAS centred)

stommel_gyre_tracer      (Hecht 2000)
acc_channel / acc_channel_rest (Zhang et al. 2024 inspired)

```

Custom IC: write a function returning `OceanState` for your grid in `experiments/` or a new module.

17 · Invalid / unwired combinations

These leaves *cannot* be selected together today:

If you pick ...	Then you cannot also pick ...	Reason
MPAS	GM/Redi triads	Phase 5 of plan, not implemented
MPAS	WENO momentum advection	only on lat-lon today
Cubed-sphere	anything multi-year	face-boundary instability
Cubed-sphere	regional / Mercator	needs 6-face init adaptation
Spectral (T21)	the main test matrix	standalone runner only

If you pick ...	Then you cannot also pick ...	Reason
<code>nlev=1</code> (truly barotropic)	KPP / convection / GM-Redi	no interior to mix
Lat-lon	implicit-CN barotropic solver	MPAS-only path today
MPAS + ETOPO	multi-year integration	algorithm-class instability
Implicit-CN baro	high <code>MAXVEL</code> dependence	clip not needed by design
GM/Redi triads	non- z^* vertical	no other coord supported

18 · Three reference trajectories from the test matrix

A. MPAS global wind-driven, multi-year stable — `global_barotropic_wind` on ico3, 60 days; with `barotropic_solver=implicit_cn` extends to 5 years. AL81 PV-flux, density-Jacobian PGF, DST-3 tracer, quadratic drag, no GM/Redi, KPP off.

B. Lat-lon Eady channel with GM/Redi triads — `eady_gm_redi` on 20×10 channel, 30 days. Centred PGF, Hollingsworth-corrected KE gradient, SOM tracer (low spurious mixing), GM+Redi with triad slopes, no KPP (idealized).

C. Lat-lon ACC channel — `acc_channel` on 20×18 channel @ 100 km, 30 days. Wind + sponges + Gaussian ridge. Vector-invariant momentum, DST-3 tracer, Smagorinsky viscosity, quadratic drag, no GM/Redi (eddy-permitting).

19 · The questions to actually ask Alistair

1. **Modal split** — should lat-lon move to implicit-CN like MPAS? Or is there a way to make the explicit + cosine path stable for the lat-lon barotropic-mode noise?
2. **Cubed-sphere face boundaries** — has he seen anything like our face-boundary instability in MITgcm cube experiments? Recipe for fix?
3. **MPAS + ETOPO** — algorithm-class limit, or fixable PGF/bathymetry-interaction bug?
4. **GM/Redi triads on Voronoi** — anyone done it correctly? (We don't have a clear precedent.)
5. **ALE adoption** — Adcroft & Hallberg perspective: should legoESM move to ALE remap on top of z^* ? What does it buy?
6. **Tracer transport in vanishing layers** — what's MOM6 actually doing today?
7. **Vertical-coordinate roadmap** — is hybrid (Bleck) worth the implementation cost for a model that already has differentiability?